

■ E-Waste Management

Comprehensive Environmental Guide

■ **Quick Tip:** Before recycling your old device, perform a factory reset and remove SIM/memory cards to protect your personal data.

■ Definition

Electronic waste (e-waste) encompasses any discarded electrical or electronic equipment (EEE) that has reached the end of its useful life. This includes computers, smartphones, televisions, printers, household appliances, wiring, and batteries. E-waste is the world's fastest-growing solid waste stream, with over 62 million metric tonnes generated globally in 2023.

E-waste contains both valuable materials — gold, silver, copper, palladium — and hazardous substances such as lead, mercury, cadmium, and brominated flame retardants, making responsible management critical for both resource recovery and environmental protection.

■■ Disposal Methods

- Certified e-waste recyclers: ensure processors hold R2, e-Stewards, or equivalent certification to guarantee responsible downstream handling.
- Manufacturer take-back programmes: many OEMs offer free or subsidised collection of their branded products.
- Retail drop-off: electronics retailers frequently operate in-store collection points.
- Municipal e-waste collection events held periodically by local governments.
- Donation or refurbishment: functional devices can be donated to schools, charities, or refurbishers to extend their useful life.

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■ Environmental Impact

Informal e-waste recycling — widespread in parts of Asia and Africa — involves burning circuit boards and using acid baths to extract metals, releasing toxic fumes and contaminating soil and water. Communities near informal recycling sites show elevated levels of heavy metals in blood, increasing cancer risk and developmental harm in children.

Formal recycling recovers precious metals more efficiently and prevents hazardous materials from entering the environment. Mining virgin metals for electronics is energy-intensive and destructive; recovering gold from e-waste requires up to 13 times less energy than primary mining. Extending device lifespans through repair and refurbishment is the single most effective strategy to reduce e-waste.

■ Safety Guidelines

- Never dismantle lithium-ion batteries yourself — damaged cells can combust spontaneously.
- Wipe all personal data before disposal: perform factory resets and use data-wiping software for computers.
- Do not place e-waste in kerbside bins or general waste — it is illegal in many jurisdictions and harmful to waste workers.
- Handle CRT monitors with care; they contain lead glass and can implode if dropped.
- Wear gloves when handling old circuit boards, as solder may contain lead.

■ Best Practices

- Buy durable, repairable devices; check repairability scores before purchasing.
- Repair rather than replace: seek authorised or independent repair services for broken devices.
- Sell or donate working devices to extend their lifecycle and offset demand for new production.
- Keep devices updated with software to prolong usability and avoid premature obsolescence.
- Advocate for right-to-repair legislation that makes spare parts and repair manuals more accessible.